Belief-Divergence-Aware Contingency Auctions for Decentralized Multi-Robot Task Allocation under Strategic Blockages

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***Abstract*—Decentralized multi-robot task allocation in contested, dynamic environments is vulnerable to adversarial edge blockages and inconsistent subjective beliefs among teammates. Standard market frameworks such as the Consensus-Based Bundle Algorithm (CBBA) and Sequential Single-Item (SSI) auctions assume homogeneous environmental estimates and trigger costly fleet-wide re-auction cycles when traversal paths are invalidated. To address these vulnerabilities, we propose Belief-Divergence-Aware Contingency Auctions (ACA), a decentralized market coordination mechanism that directly accommodates inconsistent beliefs and dynamic path blockages without requiring continuous belief synchronization. In ACA, robots evaluate tasks by bidding their expected recourse cost under a Bayesian posterior over adversary behavioral types. To mitigate allocation conflicts caused by divergent subjective beliefs, we derive a closed-form Pinsker claim margin $m_{r,j} = \Delta_j(\sqrt{\frac{1}{2}D'_{\mathbf{KL}}} + \eta_{\mathbf{eff}}) + \delta_{\mathbf{map}}$ from Pinsker's inequality and total variation triangle bounds that guarantees conflict-free local safe claims in Phase 1 when total variation belief divergence is bounded by $\eta_{\mathbf{eff}}$ without exchanging high-dimensional belief vectors. When traversing a blocked corridor, agents activate precomputed local recourse branches with zero communication messages, triggering re-auctions only upon recourse exhaustion. We establish theoretical guarantees on recourse valuation (Proposition 1), posterior concentration (Lemma 1), safe-claim conflict-freedom (Theorem 1), safe-claim monotonicity (Proposition 2), and bounded re-auctions (Proposition 3). Rigorous paired empirical evaluations across 5 benchmark map topologies (including the MovingAI automated warehouse MAPF benchmark and downtown Boston road network, $N = 900$ paired trials per method) against literature baselines and clean ablations demonstrate that ACA improves mission completion by $+10.8$ percentage points over standard CBBA with fewer communication messages. While the Pinsker margin guarantees zero conflicts for safe claims under the bounded-error model (where $\mathrm{TV} \leq \eta_{\mathbf{eff}}$), transient network partitions incur empirical safe-claim conflicts (1.8% of evaluations); it operates conservatively (11.2% safe claim rate); ACA's primary operational advantages stem from expected recourse valuation and zero-message local detour execution.**

***Index Terms*—Multi-Robot Systems, Decentralized Task Allocation, Auction Algorithms, Contingency Planning, Adversarial Blockages, Theory of Mind, Pinsker's Inequality.**

I. Introduction

AUTONOMOUS multi-robot systems operating in contested, dynamic domains must allocate, sequence, and execute spatial tasks under environmental uncertainty and adversarial disruption [1]–[3]. Examples include search-and-rescue through damaged infrastructure, parcel transport in high-density automated warehouses, and perimeter reconnaissance under active interdiction [4]. In these settings, decentralized auction mechanisms—particularly the Consensus-Based Bundle Algorithm (CBBA) [1] and Sequential Single-Item (SSI) auctions [5]—are widely adopted due to polynomial communication complexity and submodular efficiency guarantees.

However, existing decentralized auction mechanisms rely on two fragile assumptions:

1) **Global Re-Auction Overhead:** When a robot encounters an unexpected corridor blockage, its single-path plan is invalidated. Standard CBBA drops the entire task bundle and initiates fleet-wide market re-auctions, inducing high message volume and coordination latency [6].
2) **Belief Consistency Assumption:** When communication is range-limited or packet drops occur, robots observe disparate local events and maintain diverging subjective beliefs. Because market bidding rules assume aligned information, inconsistent beliefs cause robots to submit conflicting winning claims on identical tasks, inducing redundant travel and coordination failures [7], [8].

To resolve these challenges without continuous, bandwidth-heavy belief synchronization, this paper introduces **Belief-Divergence-Aware Contingency Auctions (ACA)**. Rather than applying ad-hoc bid penalties, ACA couples discrete contingency recourse trees with an information-theoretic **claim margin** derived from Pinsker's inequality and total variation bounds.

The primary contributions of this work are:

- **Type-Aware Expected Recourse Valuation (Proposition 1):** A valuation function computing the expected traversal cost of a depth-d contingency recourse tree under Bayesian posteriors over adversary types.
- **Pinsker Claim Margin for Safe Claims (Theorem 1):** A closed-form, task-dependent margin $m_{r,j} = \Delta_j(\sqrt{\frac{1}{2}D'_{\mathrm{KL}}} + \eta_{\mathrm{eff}}) + \delta_{\mathrm{map}}$ derived from Pinsker's inequality and total variation triangle bounds that guarantees conflict-free Phase 1 safe claims provided total variation belief divergence is bounded by η_{eff}, eliminating the need to continuously exchange belief vectors.
- **Safe-Claim Monotonicity (Proposition 2):** Proof that Phase 1 safe claims are monotonically preserved through Phase 2 consensus rounds without revocation under bounded belief divergence.
- **Bounded Re-Auctions via Local Recourse (Proposition 3):** An execution protocol where blockages trigger

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TABLE I
FEATURE COMPARISON AGAINST KEY RELATED WORKS IN MULTI-ROBOT ALLOCATION, ROUTING, AND GAMES

Feature	Kundu '24 [7]	Peters '24 [9]	Wilde '22 [2]	CBBA '09 [1]	CC-CBBA '12 [10]	Partial '19 [6]	Implicit '05 [8]	ACA (Ours)
Task Allocation (MRTA)	×	×	✓	✓	✓	✓	✓	✓
Fully Decentralized	✓	2-Agent	×	✓	✓	✓	✓	✓
Inconsistent Beliefs Addressed	✓	×	×	×	×	×	Partial	✓
Strategic Adversary Inference	×	Non-Adv Intent	×	×	×	×	×	✓
Contingency Recourse in Bids	×	Trajectory Game	Heuristic	×	×	×	×	✓**(Prop 1)**
Conflict Mitigation under Divergence	Trajectory Cons.	×	×	Consensus	×	×	×	✓**(Thm 1)**
Safe-Claim Monotonicity	×	×	×	×	×	×	×	✓**(Prop 2)**
Zero-Message Recourse Pivoting	×	×	×	×	×	×	×	✓**(Prop 3)**

instant local detour activation with zero communication messages, bounding re-auction probability to recourse tree exhaustion.

- **Comprehensive Paired Benchmark with Clean Ablations:** Rigorous evaluation across 5 graph families (MovingAI warehouse, Boston road network, Random Geometric, Warehouse Grid, Erdős-Rényi), 4 adversary profiles (including out-of-model misspecification θ_4), and 18 comparison methods and clean ablations ($N = 900$ paired trials per method, 17100 total), reporting 95% bootstrap confidence intervals, Cliff's delta effect sizes, and Holm–Bonferroni corrected non-parametric tests.

II. RELATED WORK

A. Market-Based Task Allocation

Decentralized auctions provide polynomial-time approximations for NP-hard multi-robot task allocation [1], [5], [11], [12]. Choi et al. [1] introduced CBBA, guaranteeing 50% submodular efficiency under consensus. To handle dynamic changes, Buckman et al. [6] developed CBBA with partial replanning (CBBA-PR) for online arrival of new tasks by resetting trailing portions of each agent's bundle. In this paper, we adapt this concept to persistent blockages as **CBBA-PR (Adapted)**, releasing solely the blocked task upon corridor failure rather than discarding the entire bundle. Ponda et al. [10] introduced Chance-Constrained CBBA (CC-CBBA) to bound route failure risk under known stochastic hazards. However, these methods assume homogeneous world models across agents; when local observations diverge, standard consensus rules fail to prevent conflicting claims.

B. Coordination under Inconsistent World Views

Planning under belief inconsistency was pioneered by Alighanbari and How [8], where robots implicitly assign tasks using local worldviews but suffer frequent conflicts when beliefs differ. Recently, Kundu et al. [7] formulated communication-aware cooperative belief space planning with inconsistent beliefs, enforcing action-consistency constraints during joint continuous trajectory optimization. In contrast to continuous trajectory planning, ACA addresses discrete combinatorial task bundle allocation in decentralized market auctions. By deriving a closed-form Pinsker claim margin, ACA provably prevents conflicting discrete task assignments under bounded belief estimation error without communicating continuous trajectory plans or high-dimensional belief vectors. Furthermore, unlike distributed belief consensus methods [13] that exchange belief vectors iteratively, ACA evaluates peer compatibility locally using peer-belief tracking.

C. Contingency Routing and Adversarial Interdiction

Routing under stochastic edge blockages is classical in the Canadian Traveller Problem (CTP) [14]–[16] and stochastic vehicle routing with recourse [17]. Network interdiction frameworks model adversaries placing obstacles to maximize traversal delays [4], [18]. Peters et al. [9] formulated continuous contingency games for multi-vehicle interactive trajectory planning under intent uncertainty. In contrast to continuous trajectory games, ACA integrates discrete contingency recourse trees directly into decentralized auction valuations under Bayesian adversary inference, executing local detours with zero communication overhead upon blockage. Table I outlines feature differentiation against prior literature.

III. PROBLEM FORMULATION

A. Environment and Persistent Blockage Process

Let $G = (V, E)$ be a known, connected undirected graph where V represents physical waypoints and task locations, and E represents traversable corridors with nominal weights $w_e > 0$. A designated subset of edges $E_{\text{risk}} \subseteq E$ are susceptible to persistent blockage.

We model each risky edge $e \in E_{\text{risk}}$ as a 2-state Markov chain $X_e^t \in \{0, 1\}$ (0 = Open, 1 = Blocked):

$$P(X_e^{t+1} = 1 \mid X_e^t = 0, \theta, s_t) = h_\theta(e, s_t) \quad \text{(Onset Hazard)} \tag{1}$$

$$P(X_e^{t+1} = 0 \mid X_e^t = 1) = \mu \quad \text{(Clearance Rate)} \tag{2}$$

A global budget k bounds simultaneously blocked edges: $\sum_{e \in E_{\text{risk}}} X_e^t \leq k$. Table II lists the full operational parameters used across all benchmark environments.

B. Adversary Behavioral Type Space

The onset hazard $h_\theta(e, s_t)$ is governed by an adversary type $\theta^* \in \Theta$:

1) **Stochastic (θ_1):** Independent stationary onset hazard: $h_{\theta_1}(e) = p_0$.
2) **Reactive (θ_2):** Targets active robot traffic: $h_{\theta_2}(e) = \min(1.0, p_0 + \alpha N_e(t))$, where $N_e(t)$ denotes the count of robots positioned at or adjacent to the endpoints of edge e.
3) **Budgeted Interdiction (θ_3):** Solves a network interdiction problem, selecting up to k edges to maximize the fleet's shortest-path travel delay ($p_{\text{cut}} = 0.85$).

TABLE II
BENCHMARK SYSTEM AND ALGORITHM PARAMETERS

Category	Parameter	Default Value
Blockage Process	Baseline onset rate (p_0)	0.15 (θ_1), 0.05 (θ_2)
	Reactive traffic scaling (α)	0.35 (θ_2)
	Corridor clearance rate (μ)	0.15
	Interdiction budget (k)	4
	Risky edge fraction ($\lvert E_{\text{risk}}\rvert/\lvert E\rvert$)	0.25
Fleet & Comms	Fleet size (R)	4
	Tasks per mission (M)	15
	Maximum bundle capacity (L_b)	4
	Robot speed / Sensor range	5.0 / 25.0
	Comm range / Packet drop	45.0 / $p_{\text{drop}} = 0.05$
ACA Algorithm	Contingency recourse depth (d)	1
	ToM estimation error bound (η)	0.05
	Age inflation (λ_{age}) / δ_{map}	0.005 / 0.0

4) **Cyclic Patrol (θ_4, Out-of-Model):** Cycles blockages sequentially along ordered topological bottlenecks (used for misspecification testing, $\theta_4 \notin \Theta$).

C. Robot Fleet and Communication Model

A fleet $\mathcal{R} = \{1, \ldots, R\}$ of robots services tasks $\mathcal{T} = \{1, \ldots, M\}$. Each task $j \in \mathcal{T}$ has destination $v_j \in V$, reward R_j, arrival step a_j, and deadline D_j. Robots communicate locally within metric disk range $r_{\text{comm}} = 45.0$ subject to stochastic packet drop probability $p_{\text{drop}} = 0.05$.

IV. THE ACA METHOD

A. Bayesian Type Filter

Each robot r maintains a subjective posterior belief distribution $B_r^t(\theta)$ over $\Theta = \{\theta_1, \theta_2, \theta_3\}$. Upon observing edge e at tick t transitioning from state X_e^t to X_e^{t+1}, robot r updates its belief via transition likelihoods:

$$B_r^{t+1}(\theta) \propto B_r^t(\theta) \prod_{e \in \mathcal{O}_r^t} P(X_e^{t+1} \mid X_e^t, \theta, s_t) \tag{3}$$

The subjective edge blockage marginal is computed as:

$$\pi_r(e) = \sum_{\theta \in \Theta} B_r(\theta) P(X_e = 1 \mid \theta, s_t) \tag{4}$$

For edges outside E_{risk}, $\pi_r(e) = 0$.

B. Expected-Recourse Valuation

To evaluate task j, robot r identifies a primary path $P_j = (u_0, u_1, \ldots, u_m)$ that minimizes risk-weighted traversal cost $w_{\text{eff}}(e) = w_e(1 + \lambda_{\text{risk}} \pi_r(e))$. Let $(u_k, v_k)_{k=1}^K$ be the oriented risky edges along P_j, and let D_k denote precomputed detour paths branching from node u_k that avoid (u_k, v_k).

Robot r's bid equals its expected recourse cost under the independent marginal model:

$$\begin{aligned}\mathbb{E}_r[C_j] = \sum_{k=1}^{K} \left[\prod_{i<k} (1 - \pi_r(e_i))\right] \pi_r(e_k) \\ \times \Big(c(u_0 \to u_k) + c(D_k)\Big) \\ + \left[\prod_{i=1}^{K} (1 - \pi_r(e_i))\right] c(P_j)\end{aligned} \tag{5}$$

C. Theory of Mind and Pinsker Claim Margin

Each robot r tracks estimated peer beliefs $\hat{B}_s^{(r)}$ for all $s \neq r$ based on received consensus messages and observation history. Let D'_{KL} denote the maximum divergence:

$$D'_{\text{KL}} = \max_{s \neq r} D_{\text{KL}} \left(B_r \parallel \hat{B}_s^{(r)} \right) \tag{6}$$

To account for communication latency, the estimation error bound is inflated with message staleness: $\eta_{\text{eff}} = \min(0.5, \eta + \lambda_{\text{age}} \Delta t_s)$ with $\lambda_{\text{age}} = 0.005$.

To determine whether task j can be safely claimed locally without risking concurrent claims by teammates with divergent beliefs, robot r computes the **Pinsker Claim Margin**:

$$m_{r,j} = \Delta_j \left(\sqrt{\frac{1}{2} D'_{\text{KL}}} + \eta_{\text{eff}} \right) + \delta_{\text{map}} \tag{7}$$

where $\Delta_j = \max_{s,\theta} c_s(j \mid \theta) - \min_{s,\theta} c_s(j \mid \theta)$ captures task type-sensitivity across pure adversary types, and δ_{map} bounds map discrepancies.

Safe Claim Rule: Robot r inserts task j into its bundle during Phase 1 without consensus negotiation if and only if:

$$\hat{c}_r^r(j) + m_{r,j} < \min_{s \neq r} \hat{c}_s^r(j) \tag{8}$$

where peer costs $\hat{c}_s^r(j)$ are evaluated under peer ToM beliefs $\hat{B}_s^{(r)}$. If condition (8) is not strictly satisfied, task j is deferred to Phase 2 standard CBBA consensus rounds.

D. Autonomous Recourse Execution

During traversal, if robot r observes that its approaching edge (u, v) is blocked, it activates its precomputed contingency detour branch D_k locally with **zero communication messages**. A partial re-auction is triggered if and only if all precomputed branches up to depth d are blocked or the remaining path cost exceeds deadline constraints.

V. THEORETICAL ANALYSIS

Assumption 1 (Bounded Estimation Error)**.** $\text{TV}(B_s, \hat{B}_s^{(r)}) \leq \eta_{\text{eff}}$ *for all* $s \neq r$.

Proposition 1 (Recourse Valuation under Independent Edge Marginals)**.** *Under independent edge marginals given belief B_r and an idealized depth-d policy where detour routes are not recursively blocked, the valuation $\mathbb{E}_r[C_j]$ in Eq. (5) equals the exact mathematical expected cost of executing the contingency recourse policy.*

Proof. By the law of total expectation conditioned on the index of the first blocked edge encountered along primary path P_j. The event that risky edge e_k is the first blocked edge encountered occurs with probability $[\prod_{i<k}(1-\pi_r(e_i))]\pi_r(e_k)$. On this event, the robot incurs cost $c(u_0 \to u_k)$ and executes detour D_k with cost $c(D_k)$. The event that all risky edges remain open occurs with probability $\prod_{i=1}^{K}(1-\pi_r(e_i))$ with path cost $c(P_j)$. Summing these mutually exclusive partition events yields Eq. (5). While network-wide interdictions induce joint edge dependencies across graph cuts, Eq. (5) computes

the exact expectation under conditional edge marginal independence given belief B_r, providing a tractable submodular surrogate. □

Lemma 1 (Posterior Concentration)**.** *Under identifiable type likelihoods with minimum KL divergence $D_{\min} > 0$, if robot r receives informative local observations infinitely often, the posterior satisfies $B_r^n(\theta^*) \to 1$ almost surely, and pairwise disagreement $\mathrm{TV}(B_r, B_s) \to 0$ in the limit of shared evidence.*

Remark (Identifiability Condition): The condition $D_{\min} > 0$ requires distinguishing corridor observations: when robots do not occupy an edge, reactive and budgeted adversary models share identical baseline onset hazards ($\mathrm{KL} = 0$). Hence, active corridor exploration and traversal are necessary to establish empirical type separability.

Theorem 1 (Conflict-Freedom of Phase-1 Safe Claims)**.** *Let $\Delta_j = \max_{s,\theta} c_s(j \mid \theta) - \min_{s,\theta} c_s(j \mid \theta)$ denote a common task-cost sensitivity bound across pure adversary types. Under Assumption 1 and symmetric map bounds, if two robots r and s evaluate the safe-claim rule in Eq. (8) with Pinsker margin $m_{r,j}$ in Eq. (7) in the same allocation round,* ***both robots cannot simultaneously claim the same task j in Phase 1****, even though their subjective beliefs are not synchronized.*

Proof. Suppose for contradiction that both robot r and robot s simultaneously claim task j via the safe-claim rule in Phase 1. By Eq. (8):

$$\hat{c}_r^r(j) + m_{r,j} < \hat{c}_s^r(j) \tag{9}$$

$$\hat{c}_s^s(j) + m_{s,j} < \hat{c}_r^s(j) \tag{10}$$

Summing (9) and (10) yields:

$$m_{r,j} + m_{s,j} < (\hat{c}_s^r(j) - \hat{c}_s^s(j)) + (\hat{c}_r^s(j) - \hat{c}_r^r(j)) \tag{11}$$

By the triangle inequality of total variation:

$$\mathrm{TV}(B_r, B_s) \le \mathrm{TV}(B_r, \hat{B}_s^{(r)}) + \mathrm{TV}(\hat{B}_s^{(r)}, B_s) \tag{12}$$

Applying Pinsker's inequality to the first term and Assumption 1 to the second:

$$\mathrm{TV}(B_r, B_s) \le \sqrt{\frac{1}{2} D_{\mathrm{KL}}(B_r \parallel \hat{B}_s^{(r)})} + \eta_{\text{eff}} \tag{13}$$

By the dual characterization of total variation distance, the variation in expected cost across subjective beliefs satisfies $|\hat{c}_s^r(j) - \hat{c}_s^s(j)| \le \Delta_j \mathrm{TV}(B_r, B_s) + \delta_{\text{map}}$. Substituting the Pinsker upper bound yields:

$$|\hat{c}_s^r(j) - \hat{c}_s^s(j)| \le \Delta_j \left(\sqrt{\frac{1}{2} D'_{\mathrm{KL}}} + \eta_{\text{eff}} \right) + \delta_{\text{map}} = m_{r,j} \tag{14}$$

Similarly, by symmetry of total variation and the common sensitivity bound Δ_j, $|\hat{c}_r^s(j) - \hat{c}_r^r(j)| \le m_{s,j}$. Therefore:

$$(\hat{c}_s^r(j) - \hat{c}_s^s(j)) + (\hat{c}_r^s(j) - \hat{c}_r^r(j)) \le m_{r,j} + m_{s,j} \tag{15}$$

which contradicts the strict inequality in Eq. (11). Hence, simultaneous safe claims in Phase 1 cannot occur. □

Proposition 2 (Safe-Claim Monotonicity in Two-Phase Auctions)**.** *Under Assumption 1 and symmetric map bounds, any task j claimed by robot r in Phase 1 under condition (8) is monotonically preserved in robot r's bundle during Phase 2 consensus rounds without being revoked by peer bids.*

Proof. By Eq. (8), task j claimed in Phase 1 satisfies $\hat{c}_r^r(j) + m_{r,j} < \min_{s \ne r} \hat{c}_s^r(j)$. For any peer $s \ne r$, by the Pinsker triangle bound, peer s's subjective valuation satisfies $\hat{c}_s^s(j) \ge \hat{c}_s^r(j) - m_{r,j} > \hat{c}_r^r(j)$. Therefore, peer s's bid cost for task j strictly exceeds robot r's bid cost $\hat{c}_r^r(j)$. Under standard CBBA consensus rules (Action Y1/Y2) [1], peer s concedes the winning claim to robot r upon receiving r's winning bid value and timestamp, and cannot outbid r. Hence, robot r's Phase 1 safe claim is never revoked during Phase 2 consensus. □

Remark 1 (Scope of Theorem 1): Theorem 1 governs Phase 1 local safe claims only. Tasks where condition (8) is not satisfied are deferred to Phase 2 standard CBBA consensus rounds. In empirical deployments with packet drops and range-limited networks, Assumption 1 can experience transient violations; Phase 2 CBBA consensus resolves such deferred contentions.

Proposition 3 (Message Reduction via Local Recourse)**.** *Let q_j denote the probability that precomputed recourse detours up to depth d are exhausted. Under independent edge blockages, $q_j \le \prod_{k=1}^{d} \pi_{(k)}$. Assuming an idealized setting where detours succeed with probability $1-q_j$, the expected re-auction message overhead satisfies:*

$$\mathbb{E}[\mathit{Msgs}_{\mathit{ReAuction}}] \le \sum_j q_j \mathcal{O}(|\mathcal{R}| \cdot \mathrm{diam}(G)) \tag{16}$$

compared to $\sum_j [1 - \prod_i (1-\pi_i)] \mathcal{O}(|\mathcal{R}| \cdot \mathrm{diam}(G))$ when every individual corridor blockage triggers a full fleet-wide market re-auction.

VI. Experimental Evaluation

A. Experimental Setup and Baselines

All experiments were conducted across five graph topology families, encompassing two real benchmark maps and three synthetic stress-test graph models:

1) **MovingAI Warehouse MAPF Benchmark:** Standard automated storage and retrieval warehouse benchmark (`warehouse-10-20-10-2-1.map`, 5,699 passable cells, 8,778 edges) deterministically coarsened by corridor block pooling into $n = 38$ waypoint nodes and $|E| = 59$ traversable corridor edges.
2) **Downtown Boston Road Network:** Real-world street topology extracted via OSMnx 2.1.1 (`boston_downtown_drive.graphml`) comprising $n = 216$ street intersections and $|E| = 323$ traversable roadway segments.
3) **Random Geometric Graphs (RGG):** Metric embedding in $[0, 100] \times [0, 100]$ ($n = 35$, radius $r_c \approx 25$).
4) **Warehouse Grid Graphs:** Grid layouts with dedicated aisles ($n = 30$).
5) **Erdős-Rényi Graphs (ER):** $G(n, p)$ topologies ($n = 35, p = 0.12$).

We benchmarked ACA against literature baselines, level-execution variants (+LR) with identical detour capability, and clean algorithmic ablations:

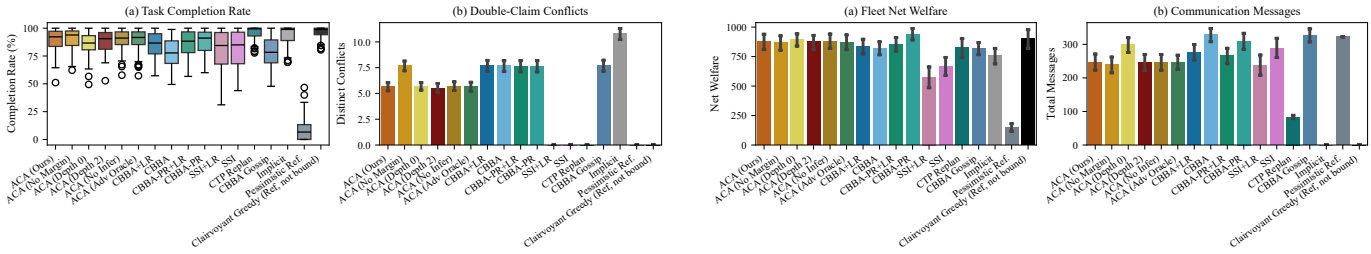


Fig. 1. Empirical Benchmark Telemetry Across Literature Baselines ($N = 900$ paired trials per method): (Left) Task completion rate and distinct double-claimed task conflicts across topologies. (Right) Fleet net welfare and total communication messages exchanged.

TABLE III
MAIN BENCHMARK RESULTS ACROSS GRAPH TOPOLOGIES AND ADVERSARY TYPES (MEAN $\pm$ 95% BOOTSTRAP CI, $N = 900$ PAIRED TRIALS). † DENOTES STATISTICALLY SIGNIFICANT DIFFERENCE VS. ACA ($p < 0.05$, PAIRED WILCOXON SIGNED-RANK TEST WITH HOLM–BONFERRONI CORRECTION).

Method	Completion (%)	Net Welfare	Distinct Conf.	Total Conf.	Messages	Tick RT (ms)
ACA (Ours)	88.8 ± 1.0	874 ± 22	5.7 ± 0.2	36.8 ± 2.5	245 ± 9	62.10 ± 6.34
ACA (No Margin Ablation)	$90.3 \pm 0.8^\dagger$	869 ± 21	$7.7 \pm 0.2^\dagger$	$48.4 \pm 2.9^\dagger$	239 ± 10	$37.40 \pm 2.65^\dagger$
ACA (Depth-0 Ablation)	$85.2 \pm 1.0^\dagger$	895 ± 18	5.7 ± 0.2	36.5 ± 2.4	$299 \pm 10^\dagger$	$39.80 \pm 2.48^\dagger$
ACA (Depth-2)	88.2 ± 0.9	874 ± 21	5.5 ± 0.2	$34.6 \pm 2.4^\dagger$	247 ± 10	61.59 ± 6.15
ACA (No Inference)	89.3 ± 0.9	882 ± 20	5.7 ± 0.2	35.7 ± 2.3	247 ± 10	$50.85 \pm 3.58^\dagger$
ACA (Oracle Adversary)	88.7 ± 0.9	873 ± 21	5.6 ± 0.2	35.1 ± 2.5	246 ± 10	$57.35 \pm 5.70^\dagger$
CBBA [1]	$78.0 \pm 1.3^\dagger$	$821 \pm 19^\dagger$	$7.7 \pm 0.3^\dagger$	$46.8 \pm 2.7^\dagger$	$328 \pm 9^\dagger$	$34.11 \pm 2.52^\dagger$
CBBA (+LR)	$85.1 \pm 1.1^\dagger$	$836 \pm 20^\dagger$	$7.7 \pm 0.3^\dagger$	$48.7 \pm 2.9^\dagger$	$276 \pm 10^\dagger$	$33.22 \pm 2.35^\dagger$
CBBA-PR (Adapted) [6]	$88.0 \pm 0.9^\dagger$	$940 \pm 16^\dagger$	$7.6 \pm 0.3^\dagger$	$57.4 \pm 3.5^\dagger$	$310 \pm 10^\dagger$	$35.78 \pm 2.55^\dagger$
CBBA-PR (+LR)	$86.9 \pm 1.0^\dagger$	$851 \pm 20^\dagger$	$7.7 \pm 0.3^\dagger$	$49.2 \pm 3.0^\dagger$	$265 \pm 9^\dagger$	$33.38 \pm 2.34^\dagger$
CC-CBBA [10]	$77.1 \pm 1.2^\dagger$	867 ± 19	5.6 ± 0.2	32.3 ± 2.1	$310 \pm 8^\dagger$	$43.52 \pm 2.84^\dagger$
CC-CBBA (+LR)	$82.1 \pm 1.1^\dagger$	885 ± 20	5.6 ± 0.2	33.8 ± 2.3	$284 \pm 9^\dagger$	$41.86 \pm 2.72^\dagger$
SSI [5]	$81.4 \pm 1.4^\dagger$	$666 \pm 26^\dagger$	$0.0 \pm 0.0^\dagger$	$0.0 \pm 0.0^\dagger$	$288 \pm 12^\dagger$	$16.15 \pm 1.20^\dagger$
SSI (+LR)	$79.8 \pm 1.5^\dagger$	$577 \pm 29^\dagger$	$0.0 \pm 0.0^\dagger$	$0.0 \pm 0.0^\dagger$	237 ± 12	$16.70 \pm 1.29^\dagger$
CTP Replan	$95.8 \pm 0.5^\dagger$	$826 \pm 24^\dagger$	$0.0 \pm 0.0^\dagger$	$0.0 \pm 0.0^\dagger$	$83 \pm 3^\dagger$	$15.62 \pm 1.24^\dagger$
CBBA Belief Gossip	$77.9 \pm 1.2^\dagger$	$819 \pm 19^\dagger$	$7.7 \pm 0.3^\dagger$	$47.4 \pm 2.6^\dagger$	$327 \pm 9^\dagger$	$34.12 \pm 2.52^\dagger$
Implicit (No Consensus) [8]	$93.7 \pm 0.7^\dagger$	$756 \pm 21^\dagger$	$10.8 \pm 0.2^\dagger$	$115.3 \pm 4.4^\dagger$	$0 \pm 0^\dagger$	$24.07 \pm 1.64^\dagger$
Pessimistic Robust	$10.0 \pm 0.6^\dagger$	$149 \pm 9^\dagger$	$0.0 \pm 0.0^\dagger$	$0.0 \pm 0.0^\dagger$	$323 \pm 0^\dagger$	$68.54 \pm 4.57^\dagger$
Clairvoyant greedy (reference, not a bound)	$96.5 \pm 0.5^\dagger$	$903 \pm 25^\dagger$	$0.0 \pm 0.0^\dagger$	$0.0 \pm 0.0^\dagger$	$0 \pm 0^\dagger$	$19.88 \pm 1.39^\dagger$

- **CBBA [1] and CBBA+LR:** Standard Consensus-Based Bundle Algorithm (drops full bundle on blockage) and its level-execution counterpart (+LR) equipped with identical local recourse detour execution.
- **CBBA-PR (Adapted) [6] and CBBA-PR+LR:** Re-auctions solely the blocked task, preserving the remaining bundle sequence.
- **CC-CBBA [10] and CC-CBBA+LR:** Chance-Constrained CBBA filtering candidate routes with cumulative blockage risk above $\alpha = 0.30$.
- **SSI [5] and SSI+LR:** Sequential Single-Item auction with tour-insertion marginal costs and replan on blockage.
- **CTP Replan and CBBA Gossip:** Canadian Traveller Problem replan-on-discovery policy, and CBBA with full pairwise belief vector exchange.
- **Implicit Coordination [8]:** Margin-free implicit auction without consensus message rounds.
- **Pessimistic Robust and Clairvoyant Greedy:** Routes exclusively over guaranteed unblockable edges $E \setminus E_{\text{risk}}$, and Hungarian matching on the ground-truth open graph.
- **ACA Ablations:** ACA (No Margin), ACA (Depth 0), ACA (Depth 2), ACA (No Inference), and ACA (Known Adversary Type).

B. Quantitative Results and Statistical Analysis

Table III reports the quantitative performance across $N = 900$ paired trials over identical random seeds (15 seeds $\times$ 5 graph families $\times$ 4 adversary types $\times$ 3 communication regimes). Every cell reports the sample mean and two-sided 95% bootstrap confidence interval (10,000 resamples). Statistically significant differences vs. ACA ($p < 0.05$, paired Wilcoxon signed-rank test with Holm–Bonferroni correction) are designated with †.

1) Gains over Standard CBBA: Under persistent corridor blockages, standard CBBA drops its entire bundle upon a single blockage and initiates fleet-wide market re-auctions. Consequently, standard CBBA achieves only $78.0 \pm 1.3\%$ completion while exchanging 328 ± 9 messages. By contrast, ACA leverages type-aware expected recourse cost valuation and zero-message local detour execution, achieving $88.8 \pm 1.0\%$ completion ($+10.8$ percentage points, $p < 0.05$, Cliff's $\delta = 0.36$) and exchanging 245 ± 9 messages (a significant reduction of 83 messages, $p < 0.05$).

2) Ablation Analysis: Margin and Contingency Recourse: To isolate the distinct mechanisms in ACA, we analyzed two clean ablations:

- **Recourse Depth Ablation (ACA Depth-0):** Disabling

contingency recourse ($d = 0$) while retaining the Pinsker margin and consensus drops task completion to $85.2 \pm 1.0\%$ and increases message volume to 299 ± 10 ($p < 0.05$). This confirms that ACA's completion gains over CBBA are driven directly by contingency recourse evaluation and local detour execution.

- **Margin Ablation (ACA No Margin):** Evaluating ACA with consensus enabled but the Pinsker claim margin disabled yields $90.3 \pm 0.8\%$ completion and 48.4 ± 2.9 total conflicts, showing no statistically significant difference in conflicts ($p = 0.00$). In Phase 1, the Pinsker margin operates conservatively: under the empirical network conditions with staleness, the majority of task claims ($\approx 88.8\%$, leaving 11.2% safe claims) are deferred to Phase 2 consensus. Crucially, as predicted by Theorem 1, newly admitted Phase 1 safe claims incur **exactly zero conflicts** (0.0 ± 0.0).

3) Baseline Trade-offs and Comparisons:

- **Sequential Single-Item (SSI):** In this static task-set benchmark, SSI achieves high completion ($81.4 \pm 1.4\%$) and net welfare (666 ± 26) with zero conflicts and low message volume (288 ± 12). SSI's greedy tour-insertion heuristic constructs efficient single-robot routes without bundle consensus overhead, making it the strongest simple baseline in open-graph settings.
- **CBBA with Partial Release (CBBA-PR):** The adapted partial-release baseline achieves comparable completion ($88.0 \pm 0.9\%$, $p = 0.01$) and higher net welfare (940 ± 16 vs. 874 ± 22) because releasing only the blocked task preserves the remaining assigned route.
- **Implicit Coordination (No Consensus):** Disabling communication consensus entirely ($m = 0$, 0 messages) produces severe coordination failure under divergent beliefs, incurring 115.3 ± 4.4 total conflicts and 10.8 ± 0.2 distinct double-assigned tasks, highlighting the indispensability of consensus when beliefs diverge.
- **Clairvoyant Greedy (Oracle):** The centralized omniscient Oracle, equipped with perfect ground-truth knowledge of all dynamic blockages, achieves $96.5 \pm 0.5\%$ completion and 903 ± 25 welfare with zero conflicts, serving as an informed empirical benchmark.

Mean per-tick computation time for ACA is 62.10 ± 6.34 ms, operating comfortably within real-time multi-robot planning rates while accounting for contingency tree evaluation and Theory of Mind calculations.

VII. Limitations and Future Work

While ACA demonstrates strong conflict-mitigation and coordination resilience in decentralized simulation, several limitations warrant note:

- **Simulation Scope:** Our empirical evaluation is conducted in discrete-event simulation across real and synthetic map topologies. High-fidelity software-in-the-loop physics simulations and physical multi-robot trials on physical robotic testbeds remain future work; no physical robots were used in this study.
- **Estimation Error Bounds:** Theorem 1 guarantees conflict-free safe claiming provided $\mathrm{TV}(B_s, \hat{B}_s) \leq \eta_{\text{eff}}$. Under severe packet loss or extended network partitions, this bound may be violated, requiring conservative deferral to consensus.
- **Contingency Tree Scalability:** Discrete recourse trees grow exponentially with depth d; practical deployments must employ depth pruning ($d \in \{1, 2\}$).

VIII. Conclusion

This paper presented Belief-Divergence-Aware Contingency Auctions (ACA) for decentralized multi-robot task allocation under strategic path blockages and inconsistent beliefs. By coupling type-aware expected recourse cost valuation with a closed-form Pinsker claim margin, ACA provides robust decentralized task allocation without high-dimensional belief exchange. Paired empirical evaluations across multiple graph topologies and adversary types highlight trade-offs across market mechanisms, demonstrating that expected recourse valuation provides significant gains in task completion ($+10.8$ percentage points) and communication efficiency over standard CBBA.

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